Answer any **one** question in this Section in the spaces provided.

7 **(a)** **(i)** Explain what is meant by an *alternating current*.

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.....

..... [1]

(ii) Fig. 7.1. shows a metal wire held taut between a knife edge X and a smooth pulley P. The wire passes between opposing poles of permanent bar magnets.

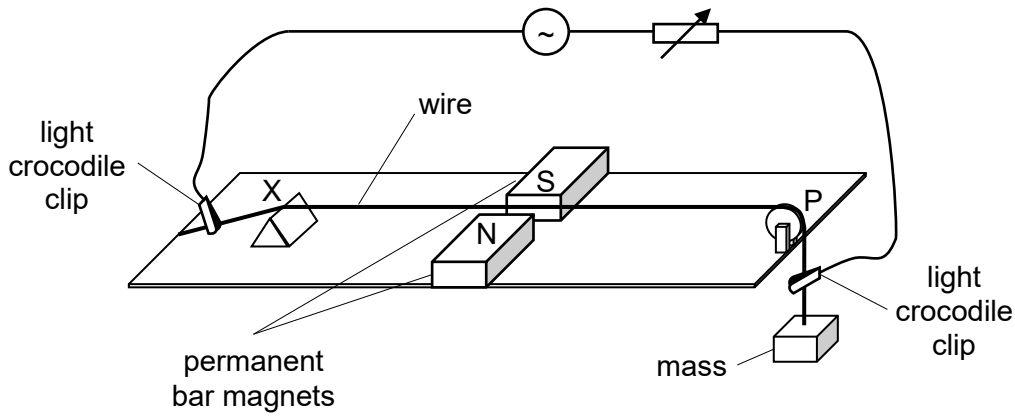

Fig. 7.1

The wire vibrates when a sinusoidal alternating source is connected across the wire. Explain how these vibrations are created. Describe the properties of these vibrations.

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..... [4]

(iii) When the frequency of the alternating source is 120 Hz, a standing wave wire in its fundamental mode is observed in the wire. XP is 2.0 m in length.

1. Calculate the speed of the wave in the wire.

speed of wave = ... m s^{-1} [2]

2. Explain, with reference to the formation of a stationary wave, what is meant by the speed calculated in **(a)(iii)1**.

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..... [4]

(iv) When the mass hanging below pulley P is doubled, the fundamental mode standing wave is observed to occur when the frequency of the alternating source is 170 Hz. When the mass was triple that of the original set up, the fundamental mode frequency is 208 Hz.

Suggest if the relationship

$$v \propto \sqrt{T}$$

is valid, where v is the wave velocity in the metal wire, and
T is the tension in the metal wire. [3]

(b) (i) State what is meant by the *diffraction* of a wave.

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..... [1]

(ii) Light of wavelength 633 nm from a laser is directed normally at a diffraction grating, as illustrated in Fig. 7.2.

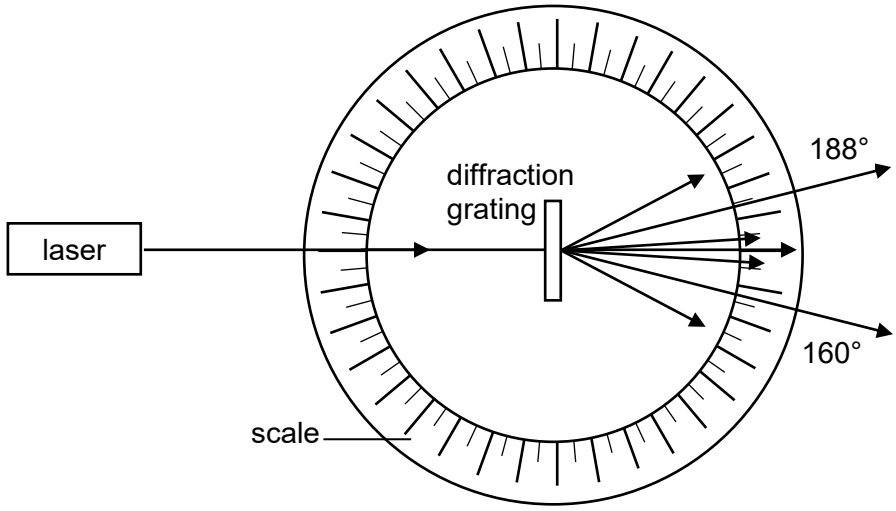

Fig. 7.2

The diffraction grating is situated at the centre of a circular scale, marked in degrees. The readings on the scale for the second order diffracted beams are 160° and 188°.

Calculate the number of lines per unit length of the slits in the diffraction grating.

number of lines per unit length = ……. m^{-1} [4]

(iii) Suggest why the non-central fringes produced by light passing through a diffraction grating is brighter than that from the same source with a double slit.

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..... [1]